

# Yang Qi

**Research Engineer** | +33 7 66 19 85 91 | qiyang0730@gmail.com | LinkedIn | Google Scholar | Homepage | GitHub | Lille, France

## Summary

Research scientist with 10+ years of experience across academia and industry, specializing in efficient machine learning architectures. Designed and implemented a neural architecture that reduced training time by 25%. Built an end-to-end PyTorch stack: tokenizer training, pretraining, SFT, response distillation, DPO, and evaluation. Seeking a research engineer role in a research-driven team.

## Core Skills

**ML / AI:** deep learning, transformer language models (LLaMA-style), pretraining, supervised fine-tuning, distillation, preference optimization (DPO), LLM evaluation, data verification, experiment design  
**Programming / Frameworks:** Python, PyTorch, TensorFlow, NumPy, scikit-learn  
**Math / Applied Research:** probability, statistics, optimization, tensor methods, signal processing, optimal control

## Experience

**Independent Researcher** | *Independent AI/ML Projects* / [GitHub \(petitgpt\)](#) | Lille, France | Jan 2025 – Present

- Built and pretrained a ~137M-parameter, 16-layer LLaMA-style (RoPE, RMSNorm, SwiGLU) decoder LM from scratch in PyTorch on a single RTX 4090.
- Ran the full post-training pipeline—supervised fine-tuning on a mixed general/code instruction dataset, targeted distillation for Python function generation, and DPO—across multiple runs, with checkpoint comparisons on an instruction-style benchmark.
- Built a code-data verification pipeline using Python AST and structural checks, unit tests in a time-bounded subprocess with allowlisted built-ins, and one round of teacher-assisted repair followed by reverification.

**Research Scientist** | *Huawei Paris Research Center* | Paris, France | Nov 2023 – Dec 2024

- Designed and implemented a neural architecture for high-dimensional CSI tensor data compression, achieving a normalized reconstruction error below 0.01 and reducing training time by 25% relative to a 3D-CNN baseline.
- Developed theory and algorithms for multi-spiked tensor recovery in high-dimensional learning, characterizing detection phase transitions and local-optimization accuracy via random matrix theory and high-dimensional optimization; derived statistical guarantees for correlated latent components and an asymptotically unbiased signal-strength estimator.

**Researcher (Inria Starting Research Position)** | *Inria and École Polytechnique* | Palaiseau, France | Nov 2019 – Oct 2023

- Initiated and developed a new research direction in data analysis, including tropical linear regression and tropical principal component analysis, which are now being applied to real-world problems in operations research.
- Developed algorithms based on tropical tensors for optimal control problems.
- Mentored two PhD students and one research intern.

**L.E. Dickson Instructor** | *University of Chicago, Dept. of Mathematics* | Chicago, IL, USA | Sep 2018 – Aug 2019

- Conducted research on neural networks and numerical analysis; mentored two master’s students.

**Postdoctoral Researcher** | *University of Chicago, Dept. of Statistics* | Chicago, IL, USA | Oct 2016 – Aug 2018

- Conducted theoretical and algorithmic research on tensor methods arising in high-dimensional statistics and optimization.

**Postdoctoral Researcher** | *CNRS / Université Grenoble Alpes / GIPSA-lab* | Grenoble, France | Oct 2013 – Sep 2016

- Conducted theoretical and algorithmic research in tensor signal processing.

## Education

**Ph.D. in Mathematics** | Texas A&M University | 2013  
**Master’s Degree in Mathematics** | Peking University | 2007

## Languages

English (fluent) | French (beginner) | Chinese (native)